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1840–1880

Railway Time: How British Towns Came to Keep London's Hour

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In the autumn of 1840 a traveller who left Paddington on the Great Western Railway carried two times with him. The timetable in his pocket gave the hours of London; the church clock at Bath or Bristol, when he arrived, gave the hours of the place, a few minutes behind. Bristol stood a little over two and a half degrees west of Greenwich, and its noon, the moment the sun crossed the local meridian, came about ten minutes after London's. For centuries the difference had not mattered. A stagecoach that took a day to cover the distance lost more than ten minutes to a lame horse. A train that covered it in four hours could not.[1]

This essay follows the forty years in which that difference was settled, from the Great Western's first timetables to the act of Parliament that made Greenwich time the legal time of Great Britain in 1880. The story has usually been

[1] For the railway's effect on the sense of time and distance, see Wolfgang Schivelbusch, *The Railway Journey: The Industrialization of Time and Space in the 19th Century* (University of California Press, 1986), chap. 3.

told as a triumph of engineering over custom. It reads better as a negotiation, in which railway companies, telegraph companies, town councils and courts each settled the question for their own purposes, and not always in the same direction.

THE COMPANIES DECIDE

The first railways ran their trains by the clocks of their own stations. On a line of a few miles this caused no trouble, and the early timetables did not say which time they meant. The difficulty came with the long trunk lines of the late 1830s, which joined towns whose clocks disagreed by a quarter of an hour.[2]

The Great Western printed London time in its timetables from November 1840 and asked its station clocks to keep it.[3] In 1847 the Railway Clearing House, the body through which the companies settled their accounts for through traffic, recommended that every member adopt Greenwich time from the first of December. By the end of 1848 nearly all of them had.[4]

The decision was commercial before it was scientific. A company that kept one time along its line needed one timetable and one set of instructions for its staff, and it could publish connections with other lines without a table of corrections. Greenwich was chosen because London was where the companies had their offices and the Post Office its mail coaches, not because the observatory stood there.[5]

THE WIRE CARRIES THE HOUR

A time that every station is meant to keep has to reach every station. At first it travelled by hand: a guard on the morning down train carried a watch set at Paddington, and the stationmasters along the line set their clocks from it.

The electric telegraph changed the means. From 1852 the Royal Observatory sent a signal each day along the wires of the Electric Telegraph Company, and the company passed it on to the railway stations and post offices on its network.[6] A clock in Manchester or Edinburgh could now be checked against

[2] Jack Simmons, *The Victorian Railway* (Thames and Hudson, 1991), 345.

[3] Derek Howse, *Greenwich Time and the Discovery of the Longitude* (Oxford University Press, 1980), 87.

[4] Howse, *Greenwich Time and the Discovery of the Longitude*, 90–91.

[5] The Post Office had sent its guards out with watches set to London time since the 1790s; see Howse, *Greenwich Time and the Discovery of the Longitude*, 82–84, on the mail-coach timepieces.

[6] Tom Standage, *The Victorian Internet* (Walker, 1998), 59.

Greenwich within a second or two of the event. The signal made Greenwich time available everywhere the wires went; it did not oblige anyone to keep it.[7]

THE TOWNS HESITATE

Most towns on a main line gave way within a few years, since a clock that disagreed with the station made people miss trains. Some held out. In 1852 the clock on the Corn Exchange at Bristol was given a second minute hand, one for Bristol and one for London, and it still has both.[8] At Exeter, which the railway reached in 1844, the cathedral clock went on keeping the time of the city for some years after the station clock had changed.[9]

The holdouts were not cranks. Local mean time followed the sun, and the sun still ordered the working day of farmers, fishermen and anyone whose hours depended on daylight. An hour fixed in London was an imposition on them. The historian of factory discipline has argued that the clock had already become a means of controlling labour by the eighteenth century;[10] the dispute over railway time extended that argument to whole towns, which were asked to accept an hour set elsewhere.

THE COURTS DECLINE

The law was slower than the towns. Statutes that named a time meant the time of the place, and that time was local mean time. In 1858 a trial at Dorchester began at ten by the court's clock, which kept London time, and judgment was given against a defendant who arrived at ten by the time of the town, some minutes later. On appeal the judges held that the hour meant was the local one, and the judgment was set aside.[11] For twenty more years a contract, a polling station or a public house could be bound by one time while the trains ran by another.

The Statutes (Definition of Time) Act of 1880 closed the gap. It declared that any expression of time in an act of Parliament, deed or other legal instrument

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[7] Schivelbusch, *The Railway Journey: The Industrialization of Time and Space in the 19th Century*, 43.

[8] Simmons, *The Victorian Railway*, 346.

[9] On the British towns, and on the same quarrel in American towns thirty years later, see Simmons, *The Victorian Railway*, 346–47; and Ian R. Bartky, *Selling the True Time: Nineteenth-Century Timekeeping in America* (Stanford University Press, 2000), chap. 9.

[10] E. P. Thompson, "Time, Work-Discipline, and Industrial Capitalism," *Past & Present*, no. 38 (1967): 80–82.

[11] Howse, *Greenwich Time and the Discovery of the Longitude*, 113.

meant, unless stated otherwise, Greenwich mean time in Great Britain and Dublin mean time in Ireland.[12]

A SETTLEMENT, NOT A CONQUEST

Seen from 1840, railway time was one of several possible arrangements. The companies might have kept local times and printed tables of corrections, as some American railroads did until 1883,[13] and the international system of zones that followed the Washington conference of 1884 took half a century to reach most of the world.[14] In Britain the distances were short, the companies were few enough to agree through a single clearing house, and the telegraph reached the stations as the trains did. Those conditions, more than any principle, account for the speed with which a country of local noons came to keep one hour.[15]

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[12] Howse, *Greenwich Time and the Discovery of the Longitude*, 114.

[13] Bartky, *Selling the True Time: Nineteenth-Century Timekeeping in America*, 140–45.

[14] Vanessa Ogle, *The Global Transformation of Time, 1870–1950* (Harvard University Press, 2015).

[15] Ogle, *The Global Transformation of Time, 1870–1950*, 25; see also Schivelbusch, *The Railway Journey: The Industrialization of Time and Space in the 19th Century*, chap. 3.